
Question #1 of 120

Question ID: 1212045

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Courtney Johnson, CFA, manages equity accounts and recommends Reliable Management to clients who ask about fixed-income investments. Reliable, in turn, provides Johnson with equity research. Johnson has not informed her equity clients, who are always very happy with Reliable's performance, of the arrangement with Reliable. Johnson has violated:

- A) none of the Standards.
- B) the Standard concerning client referrals.**
- C) the Standard concerning soft dollar arrangements.

Explanation

Standard VI(C) Referral Fees requires members and candidates to disclose to clients and prospects any consideration or benefit received by the member for the recommendation of any service. "Soft dollars" refers to benefits received from client brokerage.

For Further Reference:

(Study Session 1, 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #2 of 120

Question ID: 1212037

Russell Finley, CFA, is a managing director at Wilson Brothers and is responsible for the supervision of all trading and sales operations. Finley receives information indicating that a sales assistant made personal trades on a restricted security. According to the Standard regarding responsibilities of supervisors, the *least appropriate* action for Finley to take is to:

- A) begin an investigation to determine the extent of the wrongdoing.
- B) restrict and increase the monitoring of the employee's activities at the firm.
- C) speak directly to the employee and attain assurance that the violation will not be repeated.**

Explanation

Standard IV(C) Responsibilities of Supervisors explicitly states that speaking to the employee to determine the extent of the violations and receiving assurances that it will not be repeated is not enough. Finley must take positive steps to ensure that the violation will not be

repeated, including promptly launching an investigation and limiting the employee's activities and/or increasing supervision of the employee until the results of the investigation are known.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #3 of 120

Question ID: 1212032

Charlotte Stein, a CFA candidate, received a copy of a stock selection model designed by a Wall Street analyst friend, who told her she was free to use it. After reviewing the program and making some adjustments, Stein shows the new model to her supervisor. Her supervisor says she did a great job and tells Stein to incorporate the new model in her next industry review. Stein has:

- A) violated the Standard concerning misrepresentation.**
- B) violated the Standard concerning conflicts of interest.**
- C) not violated CFA Institute Standards of Professional Conduct.**

Explanation

Stein violated Standard I(C) Misrepresentation by presenting material developed by another to her supervisor without disclosing that the work was not her own.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #4 of 120

Question ID: 1212042

Justin Matthews, CFA, is chief financial officer of a bank and serves on the bank's investment committee. The majority of the committee has voted to invest in medium-term euro debt. Matthews feels very strongly that this is a poor strategy and that trends in both the exchange rate and in euro interest rates over the next year will result in large losses on the position. According to the Code and Standards, Matthews should *most appropriately*:

- A) document his difference of opinion with the committee.**
- B) express his concerns to the bank's chief executive officer directly.**
- C) dissociate from the recommendation by asking that his name not be included.**

Explanation

Standard V(A) Diligence and Reasonable Basis states that if a consensus opinion has a reasonable basis, a member or candidate who disagrees with it does not have to dissociate from it but should document the difference of opinion.

For Further Reference:

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #5 of 120

Question ID: 1212038

Howard Klein, CFA, supervises a group of research analysts, none of whom is a CFA charterholder or CFA candidate. He has attempted on several occasions to get his firm to adopt a compliance system to ensure that applicable laws and regulations are followed. The firm's principals, however, have never adopted his recommendations. According to CFA Institute Standards of Professional Conduct, Klein at this point:

- A) should decline in writing to accept supervisory responsibility until his firm adopts reasonable compliance procedures.**
- B) needs to take no action because the employees are not CFA charterholders or CFA candidates.**
- C) must resign from the company and document in writing his reasons for doing so.**

Explanation

Under Standard IV(C) Responsibilities of Supervisors, if Klein clearly cannot discharge supervisory responsibilities because of an inadequate compliance system, he should decline in writing to accept the supervisory responsibility until the firm adopts reasonable procedures to allow him to adequately exercise such responsibility.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #6 of 120

Question ID: 1212034

Lisa Crocker, CFA, manages several pension accounts and directs most of her trades to Zeta Brokers, which provides excellent trade execution as well as equities research. Regional Brokers, which also has excellent trading services, has offered to execute trades for Crocker at half the commission rate she pays Zeta, but Regional does not supply equities research. If Crocker declines to switch her business from Zeta to Regional, has she violated any CFA Institute Standards of Professional Conduct?

- A) Yes, because she has not obtained explicit permission from her clients to use Zeta.**
- B) No, if the higher commissions are justified by the value of the research services she receives.**
- C) Yes, because the Standard concerning loyalty, prudence, and care states that she must minimize trading costs for her accounts.**

Explanation

According to Standard III(A) Loyalty, Prudence, and Care, Crocker may pay higher fees without violating her fiduciary duty as long as the research benefits the firm's clients and the commission paid is reasonable in relation to the research and execution of services received.

For Further Reference:

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #7 of 120

Question ID: 1212039

Katrina Anderson, CFA, left her job as an account manager at RTJ Capital Management and joined Parnell Associates. Anderson did not sign a noncompete agreement at RTJ and took no RTJ property with her when she left. According to CFA Institute Standards of Professional Conduct, Anderson:

- A) must not harm RTJ by soliciting her previous clients.
- B) is free to contact her previous clients at RTJ after her employment there ends.**
- C) must seek permission from RTJ before contacting her previous clients there.

Explanation

Standard IV(A) Loyalty does not prohibit former employees from contacting clients of their previous firm so long as the contact information does not come from the records of the previous employer or violate a noncompete agreement.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #8 of 120

Question ID: 1212033

Scott Houser, CFA, is a widely known equity analyst whose recommendations often influence share prices. Houser changes his recommendation to "Sell" on Drywall Company and distributes this recommendation only to his clients, many of whom act on the recommendation before it becomes known to the public. Has Houser violated the Code and Standards?

- A) No.**
- B) Yes, he has violated the Standard concerning communications with clients.
- C) Yes, he has violated the Standard concerning material nonpublic information.

Explanation

Houser has not violated the Code and Standards. Guidance for Standard II(A) Material Nonpublic Information states that an analyst does not need to make his recommendations public just because investors would want to know about them, and is free to issue recommendations only to his clients. The guideline about what makes information material (i.e., investors would want to have the information before making an investment decision) applies to nonpublic information from the issuer of a security. An analyst covering the security, however, is not an insider with the issuing firm.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #9 of 120

Question ID: 1212046

Isabelle Burns, CFA, is an investment advisor and holds shares of Torex in her personal account because she thinks it is undervalued. According to the CFA Institute Standards of Professional Conduct, Burns may:

- A) recommend Torex to clients but must disclose her investment in Torex.**
- B) not recommend Torex to clients while she has a personal investment in the stock.**
- C) recommend Torex to clients for whom it is suitable without disclosing her investment in Torex.**

Explanation

To comply with Standard VI(A) Disclosure of Conflicts, members and candidates must make full disclosure of all matters that could impair their independence. Sell-side members and candidates should disclose to their clients any ownership in a security that they are recommending.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #10 of 120

Question ID: 1212040

Christopher Kim, CFA, is a banker with Batts Brothers, an investment banking firm. Kim follows the energy industry and has frequent contact with industry executives. Kim is contacted by the CEO of a large oil and gas corporation who wants Batts Brothers to underwrite a secondary offering of the company's stock. The CEO offers Kim the opportunity to fly on his private jet to his ranch in Texas for an exotic game hunting expedition if Kim's firm can complete the underwriting within 90 days. According to CFA Institute Standards of Conduct, Kim:

- A) may accept the offer as long as he discloses the offer to Batts Brothers.**
- B) may not accept the offer because it is considered lavish entertainment.**
- C) must obtain written consent from Batts Brothers before accepting the offer.**

Explanation

According to Standard IV(B) Additional Compensation Arrangements, members and candidates must obtain written permission from their employer before accepting an offer of compensation (for the performance of work done for their employer) in addition to what they receive from their employer and that is contingent on future performance.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #11 of 120

Question ID: 1207519

When GIPS and local laws conflict, in order to be in compliance with GIPS, the investment firm must follow:

- A) local law but disclose the conflict with GIPS.**
- B) local law, and no additional disclosure is required.**
- C) GIPS but disclose that this is in conflict with local laws.**

Explanation

Compliance requires that the firm follow local law and disclose the conflict between local law and GIPS.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.c)

CFA® Program Curriculum, Volume 1, page 237

Question #12 of 120

Question ID: 1212036

Kim Vance, CFA, tells a prospective client, "Over the three years I have been in the business, my equity-oriented accounts have had a mean return of more than 20% a year." The statement is accurate, but the mean return was influenced by the account of one client realizing a large gain on a position in a small-cap company he took based on his own research. Without this account, the average gain would have been 18% per year. Has Vance violated CFA Institute Standards of Professional Conduct?

- A) Yes, because the statement misrepresents Vance's performance.**
- B) Yes, because returns for an equities composite must be asset-weighted.**
- C) No, because it is accurate and Vance has not guaranteed such returns in the future.**

Explanation

Standard III(D) Performance Presentation requires that statements about performance be not only accurate but also fair and complete. While Vance's statement may be accurate in a technical sense, it is neither fair nor complete, and it seems intended to mislead prospects about Vance's past performance in managing equities accounts or selecting equity securities [and, thus, also likely violates Standard I(C) Misrepresentation]. While compliance with GIPS and its methods of composite construction and calculation are recommended, they are not required by the Standards.

For Further Reference:

(Study Session 1, Modules 3.2, 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #13 of 120

Question ID: 1212043

Joseph Drake, CFA, an investment advisor at Best Wealth Managers, has identified a growth stock that he believes has the potential to provide excellent returns over the next five years. He includes this stock on a "recommended list" that he sends to all of his clients. Drake includes recent earnings, his estimates of future earnings, and a note that more information is available on request. Drake has:

- A) not violated the Standards.**
- B) violated the Standard on suitability.**
- C) violated the Standard on client communications.**

Explanation

It is not a violation of the Standards to present a recommended list of securities, some of which may not be suitable for some clients. It is permitted to present investment recommendations in capsule form as long as clients are informed that more information about the securities is available on request.

For Further Reference:

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #14 of 120

Question ID: 1212047

Brian Farley, CFA, is an investment manager with one client, a \$75 million university endowment fund. A representative of the endowment fund calls Farley and places a "sell" order on a portfolio holding whose management has just reduced its earnings guidance for the coming year. Farley also owns the security and, because the new guidance is public information, places simultaneous "sell" orders for both the client account and his personal account. According to the Standards on fair dealing and priority of transactions, Farley is in violation of:

- A) both of these Standards.**
- B) neither of these Standards.**
- C) only one of these Standards.**

Explanation

Farley is in violation of Standard VI(B) Priority of Transactions. Client transactions must take precedence over members' or candidates' trades. He should have submitted his personal trade order only after trading for his client. Standard III(B) Fair Dealing requires that

Farley deal fairly with all clients when recommending securities or taking investment action. Since the endowment fund is Farley's only account, he has not disadvantaged any other client.

For Further Reference:

(Study Session 1, Modules 3.4, 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #15 of 120

Question ID: 1212041

Paul White, CFA, works as an analyst at an investment banking firm that also manages equity-only accounts for clients. White has agreed independently to manage a portfolio of fixed-income securities for an endowment fund for a small fee but has not informed his employer. Additionally, White's supervisor has asked him to work this weekend on a proposal for a large IPO that must be delivered on Monday morning, but White declines as he would prefer to spend the weekend with his family. Which of White's actions violate the Standard concerning loyalty?

- A) Both of these actions.
- B) Neither of these actions.
- C) **Only one of these actions.**

Explanation

The Standards do not require that members put their employment ahead of their personal lives; these are issues between White and his employer. However, Standard IV(A) Loyalty states that a member who engages in independent practice must notify his employer.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #16 of 120

Question ID: 1212035

Art Dodd, CFA, is a registered representative with Owens Securities. He is currently in a dispute with one client, Madge Phillips, about a limit order for her account that she feels was entered incorrectly, resulting in a loss (in her opinion) of \$500. Dodd has 1,000 shares of an oversubscribed new issue to allocate to clients. He suggests to Phillips that he will give her 250 shares of this allocation to make up for the supposed trade error. Further, he offers to buy her dinner at a nice restaurant. According to the Standards of Practice, Dodd has *most likely* violated the Standard concerning:

- A) misconduct.
- B) **fair dealing.**
- C) additional compensation.

Explanation

Dodd has most likely violated Standard III(B) Fair Dealing by giving Phillips an overgenerous allocation of the oversubscribed ("hot") new issue. The problem is not that he is offering compensation to a client to resolve a dispute but that by overallocating the IPO shares to Phillips, he is not treating his other clients fairly. Standard III(B) Fair Dealing requires that members and candidates not use shares of hot issues as an incentive to achieve a reward or benefit. The benefit in this case is that the dispute will be resolved. Dodd's actions do not exhibit any dishonesty that would violate Standard I(D) Misconduct, and Dodd is not receiving any additional compensation.

For Further Reference:

(Study Session 1, Modules 3.2, 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #17 of 120

Question ID: 1207520

Which of the following is *least likely* one of the nine major topics of the CFA Institute Global Investment Performance Standards (GIPS)?

- A) Real Estate.
- B) Private Equity.
- C) **Venture Capital.**

Explanation

The nine major sections of the CFA Institute Global Investment Performance Standards are Fundamentals of Compliance, Input Data, Calculation Methodology, Composite Construction, Disclosures, Presentation and Reporting, Real Estate, Private Equity, and Wrap Fee/Separately Managed Account (SMA) Portfolios.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.d)

CFA® Program Curriculum, Volume 1, page 239

Question #18 of 120

Question ID: 1212044

Graham Carson, CFA, is an investment advisor to Ron Grayson, a client with moderate risk tolerance and an investment horizon of 15 years. Grayson calls Carson to complain about two stocks in his account that have performed poorly. He feels that one stock was too risky for him as it paid no dividend and had a beta of 1.4. The other stock had a beta of 0.9 and paid a dividend of 3%, but financial regulators have indicated that the firm's reported earnings were incorrectly stated. Based on this information, Carson has *most likely*:

- A) **not violated the Standards.**
- B) violated only the Standard on suitability.
- C) violated both the Standard on suitability and the Standard on diligence and reasonable basis.

Explanation

Carson has not violated either Standard based on the information given. The suitability of an investment is to be determined based on the risk and return characteristics of the portfolio and not on the risk and return characteristics of each individual security. The fact that a security does not pay a dividend and has a beta higher than the market is not enough to determine its suitability in a portfolio context.

The fact that regulators have called previously reported earnings into question does not necessarily mean that Carson's analysis was not diligent or that he did not have a reasonable basis for his selection of this security.

For Further Reference:

(Study Session 1, Module 3.5, 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

Question #19 of 120

Question ID: 1207522

Questions 19 through 30 relate to Quantitative Methods. (18 minutes)

Five years ago, an investor borrowed \$5,000 from a financial institution that charged a 6% annual interest rate, and he immediately took his family to live in Nepal. He made no payments during the time he was away. When he returned, he agreed to repay the original loan plus the accrued interest by making five end-of-year payments starting one year after he returned. If the interest rate on the loan is held constant at 6% per year, what annual payment must the investor make in order to retire the loan?

- A) \$1,338.23.
- B) \$1,588.45.**
- C) \$1,638.23.

Explanation

With no interest paid on the original \$5,000 loan, at 6% in five years the loan balance will be:

$$\text{New loan balance} = \$5,000(1.06)^5 = \$6,691.13 \text{ or } PV = 5,000; I/Y = 6; N = 5; PMT = 0; CPT \rightarrow FV = -\$6,691.13.$$

\$6,691.13 is the loan that has to be retired over the next five years. The financial calculator solution is:

$$PV = 6,691.13; I/Y = 6; N = 5; FV = 0; CPT \rightarrow PMT. \text{ You obtain } PMT = -1,588.45.$$

For Further Reference:

(Study Session 2, Module 6.3, LOS 6.e)

CFA® Program Curriculum, Volume 1, page 316

Question #20 of 120

Question ID: 1207521

If an investment of \$4,000 will grow to \$6,520 in four years with monthly compounding, the effective annual interest rate will be *closest* to:

- A) 11.2%.
- B) 12.3%.
- C) 13.0%.**

Explanation

$$N = 4; PMT = 0; PV = -4,000; FV = 6,520; CPT \rightarrow I/Y = 12.99\%,$$

or $\left(\frac{6,520}{4,000}\right)^{1/4} - 1 = 12.99\%$

The question asks for the *effective* annual rate and gives the beginning and ending values. That the ending value was arrived at by monthly compounding is not relevant.

For Further Reference:

(Study Session 2, Module 6.1, LOS 6.c)

CFA® Program Curriculum, Volume 1, page 321

Question #21 of 120

Question ID: 1207523

An analyst constructs a histogram and frequency polygon of monthly returns for aggressive equity funds over a 20-year period. Which of the following statements about these displays is *most accurate*?

- A) The height of each bar in a frequency polygon represents the absolute frequency for each return interval.
- B) Both a histogram and a frequency polygon provide a graphical display of data found in a frequency distribution.
- C) To construct a histogram, the analyst would plot the midpoint of the return intervals on the x-axis and the absolute frequency for that interval on the y-axis, connecting neighboring points with a straight line.

Explanation

Choice A describes a histogram, and Choice C describes a frequency polygon.

For Further Reference:

(Study Session 2, Module 7.1, LOS 7.d)

CFA® Program Curriculum, Volume 1, page 381

Question #22 of 120

Question ID: 1212049

An analyst has data on institutional salespeople at an investment banking firm showing how they ranked in total monthly commissions, from first to eighth. To determine whether a high rank in one month indicates a high probability of achieving a high rank in subsequent months, the analyst should use a:

- A) *t*-test.
- B) nonparametric test.
- C) mean differences test.

Explanation

A test of the correlations between monthly ranks is appropriate. This is a nonparametric test.

For Further Reference:

Study Session 3, Modules 11.3, LOS 11.I

Question #23 of 120

Question ID: 1207524

A security has annual returns of 5%, 10%, and 15%. The coefficient of variation of the security (using the population standard deviation) is *closest* to:

- A) 0.3.
- B) 0.4.
- C) 0.5.

Explanation

$$\text{mean} = \frac{5+10+15}{3} = 10$$

$$\begin{aligned}\text{population standard deviation} &= \sqrt{\frac{(5-10)^2 + (10-10)^2 + (15-10)^2}{3}} \\ &= 4.0825\end{aligned}$$

$$\text{CV} = \frac{\text{standard deviation}}{\text{mean}} = \frac{4.0825}{10} = 0.408$$

For Further Reference:

(Study Session 2, Module 7.3, LOS 7.i)

CFA® Program Curriculum, Volume 1, page 422

Question #24 of 120

Question ID: 1207525

If an analyst concludes that the distribution of a large sample of returns is positively skewed, which of the following relationships involving the mean, median, and mode is *most likely*?

- A) Mean > median > mode.
- B) Mean < median < mode.
- C) Mean > median < mode.

Explanation

For the positively skewed distribution, the mode is less than the median, which is less than the mean. (Study Session 2, Module 7.3, LOS 7.k)

For Further Reference:

(Study Session 2, Module 7.3, LOS 7.k)

CFA® Program Curriculum, Volume 1, page 424

Question #25 of 120

Question ID: 1207526

An analyst has been hired to evaluate a high-risk project. The analyst estimates the probability that the project will fail in the first year as well as the conditional probability of failure for each of the remaining four years of the project, as follows:

Year	1	2	3	4	5
Failure probability	0.25	0.20	0.20	0.15	0.10

The project will have no payoff if it fails, but it will have a payoff of \$20,000 at the end of the fifth year if it succeeds. Because of its high risk, the required rate of return for an investment in this project is 25%. Based on this information, the expected present value of the project is *closest* to:

- A) \$2,400.
- B) \$3,010.
- C) \$5,900.

Explanation

The probability of surviving to the end of the fifth year is the product of the probabilities of surviving in each year, or one minus the probability of failure. Therefore, the 5-year survival probability is $(1 - 0.25)(1 - 0.20)(1 - 0.20)(1 - 0.15)(1 - 0.10) = 0.3672$.

The present value of the expected payoff is $\frac{\$20,000(0.3672)}{1.25^5} = \$2,407$.

For Further Reference:

(Study Session 2, Module 8.2, LOS 8.i)

CFA® Program Curriculum, Volume 1, page 474

Question #26 of 120

Question ID: 1212048

The probability that fund manager will produce returns in excess of the returns on the S&P 500 index in any one year is 45%. The probability that the manager will produce returns in excess of the returns on the S&P 500 index in 4 or 5 of the next five years is *closest* to:

- A) 11.3%.
- B) 14.4%.
- C) 16.3%.

Explanation

We need to use the binomial formula to calculate the probabilities of 4 and 5 "successes" in five 5 years and then sum them.

$$\text{Prob (4)} = \frac{5!}{(5-4)!4!} \times 0.45^4 \times 0.55^{5-4} = 11.3\%$$

$$\text{Prob (5)} = 0.45^5 = 3.1\%$$

$$11.3\% + 3.1\% = 14.4\%.$$

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.f)

Question #27 of 120

Question ID: 1207528

Which of the following distributions is *most likely* a discrete distribution?

- A) A normal distribution.
- B) A univariate distribution.
- C) **A binomial distribution.**

Explanation

The binomial distribution is a discrete distribution, while the normal distribution is an example of a continuous distribution. Univariate distributions can be discrete or continuous.

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.a)

CFA® Program Curriculum, Volume 1, page 514

Question #28 of 120

Question ID: 1207529

An investment has a mean return of 15% and a standard deviation of returns equal to 10%. If the distribution of returns is approximately normal, which of the following statements is *least accurate*? The probability of obtaining a return:

- A) less than 5% is about 16%.
- B) greater than 35% is about 2.5%.
- C) **between 5% and 25% is about 95%.**

Explanation

About 68% of all observations fall within ± 1 standard deviation of the mean. Thus, about 68% of the values fall between 5 and 25.

For Further Reference:

(Study Session 3, Module 9.2, LOS 9.I)

CFA® Program Curriculum, Volume 1, page 534

Question #29 of 120

Question ID: 1207530

Which of the following statements about the central limit theorem is *least accurate*?

- A) **The central limit theorem has limited usefulness for skewed distributions.**
- B) The mean of the population and the mean of all possible sample means are equal.
- C) When the sample size is large, the sampling distribution of the sample means is approximately normal.

Explanation

The central limit theorem holds for any distribution as long as the sample size is large (i.e., $n > 30$).

For Further Reference:

Question #30 of 120

Question ID: 1207531

Which of the following statements about hypothesis testing involving a z -statistic is *least accurate*?

- A) The p -value is the smallest significance level at which the null hypothesis can be rejected.
- B) A z -test is theoretically acceptable in place of a t -test for tests concerning a mean when sample size is small.
- C) If the confidence level is set at 95%, the chance of rejecting the null hypothesis when in fact it is true is 5%.

Explanation

The t -test must be used when the sample size is small, the population is normal, and the population variance is unknown. If the population is non-normal and the variance is unknown, there is no valid test statistic when the sample is small.

For Further Reference:

CFA® Program Curriculum, Volume 1, page 615

CFA® Program Curriculum, Volume 1, page 619

CFA® Program Curriculum, Volume 1, page 620

Question #31 of 120

Question ID: 1207541

Questions 31 through 42 relate to Economics. (18 minutes)

According to the crowding-out effect, the sale of government bonds used to finance excess government spending is *least likely* to:

- A) increase the real interest rate.
- B) reduce private investment spending.
- C) increase the profitability of corporate investment projects.

Explanation

Increased government borrowing would decrease, not increase, the profitability of corporate investment projects since it will tend to increase interest rates and required rates of return in general.

For Further Reference:

(Study Session 5, Module 16.3, LOS 16.q)

CFA® Program Curriculum, Volume 2, page 305

Question #32 of 120

Question ID: 1207542

Which of the following statements on the economic implications of trade restrictions is *most accurate*?

- A) Quota rents are the amounts received by the domestic government when it charges for import licenses.
- B) In the importing country, import quotas, tariffs, and voluntary export restraints all decrease producer surplus.
- C) **In the case of a quota, if the domestic government collects the full value of the import licenses, the result is the same as that of a tariff.**

Explanation

If the domestic government collects the full value of the import license, a quota can have the same economic result as a tariff. Quota rents are the gains to those foreign exporters who receive import licenses under a quota if the domestic government does not charge for the import licenses. With respect to the importing country, import quotas, tariffs, and voluntary export restraints all decrease consumer surplus and increase producer surplus.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.e)

CFA® Program Curriculum, Volume 2, page 352

Question #33 of 120

Question ID: 1207539

Assume that the long-term equilibrium money market interest rate is 4% and the current money market interest rate is 3%. At this current rate of 3%, there will be an excess:

- A) demand for money in the money market, and investors will tend to be net buyers of securities.
- B) **demand for money in the money market, and investors will tend to be net sellers of securities.**
- C) supply of money in the money market, and investors will tend to be net buyers of securities.

Explanation

At interest rates below 4% (the long-term equilibrium rate), the quantity of money demanded exceeds the quantity of money supplied. At below-equilibrium rates, investors will sell bonds to obtain the desired extra cash. As they sell more bonds, the prices of bonds fall, and interest rates start to move back towards the 4% equilibrium.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.d)

CFA® Program Curriculum, Volume 2, page 269

Question #34 of 120

Question ID: 1207537

Based on her forecast for the economy, a portfolio manager increases her investments in high-quality bonds and decreases her investments in commodities. The portfolio manager *most likely* expects the economy to experience:

- A) stagflation.
- B) **a recessionary gap.**
- C) an inflationary gap.

Explanation

The manager's investment decisions are most consistent with expectations of a recessionary gap. In a recession, commodity prices and interest rates are likely to decrease. Decreasing interest rates should increase the prices of high-quality bonds. An inflationary gap would likely cause interest rates to increase, which would decrease bond prices. An inflationary gap or stagflation conditions would likely result in increasing commodity prices.

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.i)

CFA® Program Curriculum, Volume 2, page 161

Question #35 of 120

Question ID: 1207538

An analyst who is interested in the core inflation rate should *most appropriately* examine a price index:

- A) for wholesale goods.
- B) that uses hedonic pricing.
- C) that excludes food and energy.

Explanation

Core inflation refers to prices excluding food and energy, which tend to exhibit higher price volatility than most other goods and services. Hedonic pricing refers to adjusting a price index for improvements in the quality of goods.

For Further Reference:

Study Session 4, Module 15.2, LOS 15.f, 15.g

CFA® Program Curriculum, Volume 2, page 225

CFA® Program Curriculum, Volume 2, page 227

Question #36 of 120

Question ID: 1207533

Demand for gasoline (in hundreds of liters) at a particular station, as a function of the price of gasoline and the price of bus travel, is $Q_D = 300 - 14 P_{\text{gas}} + 2 P_{\text{bus}}$. If the price of gasoline per liter (P_{gas}) is 1.50 euros, and the price of a standardized unit of bus travel (P_{bus}) is 12 euros, the cross price elasticity of gasoline demand with respect to the price of bus travel is *closest* to:

- A) 0.01.
- B) 0.08.
- C) 2.00.

Explanation

To calculate the cross price elasticity of the quantity demanded of gasoline with respect to the price of bus travel, we must first calculate the quantity of gas demanded:

$$300 - 14(1.5) + 2(12) = 303$$

The cross elasticity is:

$$\frac{\Delta Q_D}{\Delta P_{bus}} \times \frac{P_{bus}}{Q_D} = 2 \times \left(\frac{12}{303} \right) = 0.0792 \text{ or } 0.08$$

For Further Reference:

(Study Session 4, Module 12.1, LOS 12.a)

CFA® Program Curriculum, Volume 2, page 9

Question #37 of 120

Question ID: 1212050

Which of the following statements about a monopolist is *least accurate*?

- A) The monopolist faces a downward sloping demand curve.
- B) **Unlike an oligopolist, a monopolist will always be able to earn economic profit.**
- C) A profit-maximizing monopolist will expand output until marginal revenue equals marginal cost.

Explanation

In some cases, a monopolist may be unable to sell for a profit. Price may be insufficient to cover the per-unit cost of the monopolist, even when operating at the MR = MC rate of output. The monopolist faces a downward-sloping demand curve.

For Further Reference:

Study Session 4, Module 13.4, LOS 13.a, 13.b, 13.d

CFA® Program Curriculum, Volume 2, page 64

CFA® Program Curriculum, Volume 2, page 64

CFA® Program Curriculum, Volume 2, page 64

Question #38 of 120

Question ID: 1207535

A market has the following characteristics: a large number of independent sellers, each producing a differentiated product; low barriers to entry; producers facing downward sloping demand curves; and demand that is highly elastic. This description *most* closely describes:

- A) an oligopoly.
- B) pure competition.
- C) **monopolistic competition.**

Explanation

These conditions characterize *monopolistic competition*. By contrast, monopolies and oligopolies have high barriers to entry and involve either a single seller (monopoly) or a small number of interdependent sellers (oligopoly). Similar to monopolistic competition, pure competition involves a large number of independent sellers. With pure competition, products are homogeneous (not differentiated), no barriers to entry exist (not low barriers to entry), and the demand schedule is horizontal (not downward sloping) and perfectly elastic (not highly elastic).

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.h)

Question #39 of 120

Question ID: 1207543

One year ago, the currency of Xyland (XYZ) was at a three-month forward premium to the currency of Piqua (PQR). Today, the XYZ is at a three-month forward discount to the PQR. Assuming the interest rate forward parity relationship holds, this change implies that:

- A) the XYZ has depreciated relative to the PQR.
- B) today the XYZ three-month interest rate is higher than the PQR three-month interest rate.**
- C) one year ago the XYZ three-month interest rate was higher than the PQR three-month interest rate.

Explanation

Interest rate parity requires that the currency with the higher interest rate will sell at a discount in the forward market.

For Further Reference:

Study Session 5, Module 18.2, LOS 18.e, 18.g

CFA® Program Curriculum, Volume 2, page 421

CFA® Program Curriculum, Volume 2, page 421

Question #40 of 120

Question ID: 1207536

If domestic savings are insufficient to finance domestic private investment and exports are greater than imports, it is *most likely* that the fiscal budget has:

- A) a deficit that is less than the trade surplus.
- B) a deficit that is greater than the trade surplus.
- C) a surplus that is greater than the trade surplus.**

Explanation

The fundamental relationship among saving, investment, the fiscal balance and the trade balance is expressed as $(S - I) = (G - T) + (X - M)$. If domestic savings (S) are not sufficient to finance private investment (I), then $(S - I)$ is negative and the sum $(G - T) + (X - M)$ must also be negative. With exports greater than imports, $(X - M)$ is positive so $(G - T)$ must be negative and larger than $(X - M)$. If $(G - T)$ is negative, taxes (T) are greater than government spending (G) and the government has a fiscal surplus.

For Further Reference:

(Study Session 4, Module 14.1, LOS 14.e)

CFA® Program Curriculum, Volume 2, page 135

Question #41 of 120

Question ID: 1207544

Depreciation of a country's currency will be more effective in reducing its trade deficit if its:

- A) imports do not have good substitutes.
- B) exports are primarily luxury goods.**
- C) exports represent a small portion of foreign consumer expenditures.

Explanation

Under the elasticities approach, a currency depreciation will lead to a greater reduction in a trade deficit when export demand and/or import demand are more elastic. The demand for luxury goods is relatively elastic, while the demand for goods without good substitutes or for goods that represent only a small portion of consumer expenditures is relatively inelastic.

For Further Reference:

(Study Session 5, Module 18.3, LOS 18.j)

CFA® Program Curriculum, Volume 2, page 440

Question #42 of 120

Question ID: 1207540

Open market sales of securities by a country's central bank will *most likely* result in:

- A) decreasing short-term interest rates.
- B) appreciation of the domestic currency.**
- C) an increasing growth rate of real GDP.

Explanation

Central bank sales of securities reduce excess reserves in the banking system, causing interbank lending rates and other short-term interest rates to increase. If the monetary policy transmission mechanism operates normally, long-term interest rates should also increase, the domestic currency should appreciate, and economic growth and inflation should decrease.

For Further Reference:

(Study Session 5, Module 16.2, LOS 16.k)

CFA® Program Curriculum, Volume 2, page 283

Question #43 of 120

Question ID: 1207552

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

Normal Corp. has a current ratio above 1 and a quick ratio less than 1. Which of the following actions will increase the current ratio and decrease the quick ratio? Normal Corp.:

- A) buys fixed assets on credit.
- B) uses cash to purchase inventory.
- C) pays off accounts payable from cash.**

Explanation

Paying off accounts payable from cash lowers current assets and current liabilities by the same amount. Because the current ratio started off above 1, the current ratio will increase. Because the quick ratio started off less than 1, it will decrease further. The other choices are incorrect. Buying fixed assets on credit decreases both ratios because the denominator increases, with no change to the

numerator. Using cash to purchase inventory would result in no change in the current ratio but would decrease the quick ratio by decreasing the numerator.

For Further Reference:

(Study Session 7, Module 24.2, LOS 24.b)

CFA® Program Curriculum, Volume 3, page 256

Question #44 of 120

Question ID: 1212051

A company has 1,000,000 warrants outstanding at the beginning of the year, each convertible into one share of stock with an exercise price of \$50. No new warrants were issued during the year. The average stock price during the period was \$60, and the year-end stock price was \$45. What adjustment for these warrants should be made, under the treasury stock method, to the number of shares used to calculate diluted earnings per share (EPS)?

- A) 0.
- B) 166,667.
- C) 200,000.

Explanation

Diluted EPS uses average price. Since the average price is greater than the exercise price, the warrants are dilutive.

$$\frac{60-50}{60} \times 1,000,000 = 166,667$$

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g, 21.h)

CFA® Program Curriculum, Volume 3, page 99

CFA® Program Curriculum, Volume 3, page 99

Question #45 of 120

Question ID: 1207545

A firm has undertaken a contract with an estimated total cost of \$200 million at a price of \$220 million. At the end of the first reporting period, the firm has devoted resources of \$70 million to the project. The customer has been billed for \$80 million and made payments of \$60 million. As a result of these transactions, the firm should report revenue from this project of:

- A) \$60 million.
- B) \$70 million.
- C) \$77 million.

Explanation

Using the percentage of total costs incurred to date as an estimate of the portion of the performance obligations completed, revenue should be $(70/200) \times \$220 \text{ million} = \77 million .

For Further Reference:

(Study Session 7, Module 21.2, LOS 21.c)

Question #46 of 120

Question ID: 1207551

An analyst gathered the following data about a company:

Net sales	\$4,000
Dividends declared	170
Cost of goods sold	2,000
Inventory increased by	100
Accounts payable increased by	300
Cash expenses for other inputs	500
Long-term debt principal repayment	250
Cash tax payments	200
Purchase of new equipment	300

The company's cash flow from operations, based on these data only, is:

- A) \$1,200.
- B) **\$1,500.**
- C) \$1,575.

Explanation

Sales	+\$4,000	
Cash received from customers	+\$4,000	(since no change in AR)
Cost of goods sold	−2,000	
Increase in inventory	−100	
Increase in accounts payable	+300	
Other cash input expenses	−500	
Cash paid for inputs	−2,300	
Cash paid for taxes	−200	
Cash flow from operations	+1,500	

For Further Reference:

(Study Session 7, Module 23.2, LOS 23.f)

CFA® Program Curriculum, Volume 3, page 199

Question #47 of 120

Question ID: 1212054

A low inventory turnover ratio in a period of declining revenue growth is *most likely* an indication that a firm may have:

- A) **obsolete inventory.**
- B) too little inventory.
- C) efficient inventory management.

Explanation

Low inventory turnover and declining revenue growth may be signs that a firm has obsolete or slow-moving inventory. High turnover and low revenue growth may indicate too little inventory, while high turnover and high revenue growth may indicate efficient inventory management.

For Further Reference:

(Study Session 8, Module 25.5, LOS 25.I)

CFA® Program Curriculum, Volume 3, page 340

Question #48 of 120

Question ID: 1207554

The following data pertains to a company's common-size financial statements.

- Current assets 40%
- Total debt 40%
- Net income 16%
- Total assets \$2,000
- Sales \$1,500
- Total asset turnover ratio 0.75
- The firm has no preferred stock in its capital structure.

The company's after-tax return on common equity is *closest* to:

- A) 15%.
- B) **20%.**
- C) 25%.

Explanation

$$\text{ROE} = \frac{\text{net income}}{\text{equity}} = \frac{0.16(1,500)}{(1-0.40)(2,000)} = 0.20, \text{ or } 20\%$$

If the debt ratio (TD/TA) is equal to 40% and the firm has no preferred stock, the percentage of equity is 1 – 0.40, or 60%.

For Further Reference:

(Study Session 7, Module 24.3, LOS 24.b)

CFA® Program Curriculum, Volume 3, page 256

Question #49 of 120

Question ID: 1207546

Which of the following statements about the calculation of earnings per share (EPS) is *least accurate*?

- A) Shares issued after a stock split must be adjusted for the split.
- B) Options outstanding may have no effect on diluted EPS.
- C) Reacquired shares are excluded from the computation from the date of reacquisition.

Explanation

Shares issued post-split need not be adjusted for the split as they are already "new" shares. Options with an exercise price greater than the average share price do not affect diluted EPS.

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g)

CFA® Program Curriculum, Volume 3, page 99

Question #50 of 120

Question ID: 1207553

A company has a cash conversion cycle of 70 days. If the company's payables turnover decreases from 11 to 10 and days of sales outstanding increase by 5, the company's cash conversion cycle will:

- A) decrease by approximately 8 days.
- B) decrease by approximately 3 days.
- C) increase by approximately 2 days.

Explanation

cash conversion cycle (CCC) = days of sales outstanding + days of inventory on hand – number of days of payables

$$\text{number of days of payables} = \frac{365}{\text{payables turnover}} = \frac{365}{11} = 33.18 \text{ days;}$$
$$\frac{365}{10} = 36.5 \text{ days}$$

Since the payables payment period increases by 3.32 days and receivables days increases by 5, CCC increases by 1.68 days.

For Further Reference:

(Study Session 7, Module 24.2, LOS 24.b)

CFA® Program Curriculum, Volume 3, page 256

Question #51 of 120

Question ID: 1207558

Selected items from the financial statements of three plumbing fixture manufacturers appear in the following table. All three firms use straight-line depreciation for financial reporting.

	Robbco	Sammco	Teddco
PP&E at historical cost	\$10,000,000	\$12,000,000	\$18,000,000

Accumulated depreciation	6,000,000	8,000,000	9,000,000
Net PP&E	4,000,000	4,000,000	9,000,000
Depreciation expense (latest period)	800,000	1,000,000	1,500,000

Based only on these data, which firm is *most likely* to require significant capital expenditures sooner than the others?

- A) Robbco.
- B) Sammco.
- C) Teddco.

Explanation

Sammco has PP&E with the highest average age and lowest remaining useful life, and therefore appears more likely to need new equipment in the near future than the other firms.

Average age

Robbco: $\$6,000,000 / \$800,000 = 7.5$ years

Sammco: $\$8,000,000 / \$1,000,000 = 8.0$ years

Teddco: $\$9,000,000 / \$1,500,000 = 6.0$ years

Remaining useful life

Robbco: $\$4,000,000 / \$800,000 = 5$ years

Sammco: $\$4,000,000 / \$1,000,000 = 4$ years

Teddco: $\$9,000,000 / \$1,500,000 = 6$ years

For Further Reference:

(Study Session 8, Module 26.4, LOS 26.m)

CFA® Program Curriculum, Volume 3, page 417

Question #52 of 120

Question ID: 1207560

In accounting for a defined benefit pension plan, the amount reported as "prior service cost" refers to:

- A) the total value of benefits already paid to retirees who are still receiving pension payments.
- B) the present value of the pension benefits due to employees based on their employment up to the date of the statement.
- C) the present value of the increase in future pension benefits from a change in the terms of the pension plan.

Explanation

Prior service costs arise when changes in the terms of a defined benefit pension plan increase the future benefits due to employees based on their prior employment with the company.

For Further Reference:

Study Session 8, LOS 28.i

Question #53 of 120

Question ID: 1207559

On January 2, a company acquires some state-of-the-art production equipment at a net cost of \$14 million. For financial reporting purposes, the firm will depreciate the equipment over a 7-year life using straight-line depreciation and a zero salvage value; for tax reporting purposes, however, the firm will use straight-line depreciation over a 3-year life. Given a tax rate of 35%, by how much will the company's deferred tax liability increase in the first year of the equipment's life?

- A) \$931,700.
- B) \$1,064,800.
- C) \$1,730,300.

Explanation

Straight-line depreciation: \$14 million / 7 = \$2.0 million	
Accelerated depreciation: \$14 million / 3 = \$4.662 million	
Difference in depreciation: \$4.662 million – \$2.0 million = \$2.662 million	
× Tax rate	0.35
Increase in deferred tax liability \$931,700	

For Further Reference:

(Study Session 8, Module 27.3, LOS 27.d)
CFA® Program Curriculum, Volume 3, page 454

Question #54 of 120

Question ID: 1207561

When comparing two firms, an analyst should *most* appropriately adjust the financial statements when they include significant:

- A) acquisition goodwill, if one of the firms reports under IFRS and the other under U.S. GAAP.
- B) **property, plant, and equipment, if one of the firms uses accelerated depreciation and the other uses straight-line depreciation.**
- C) unrealized losses from securities held for trading, if one of the firms uses fair value reporting for securities investments and the other does not.

Explanation

Depreciation methods are an example of a difference that may require an analyst to adjust financial statements to make them comparable. Acquisition goodwill is treated the same way under IFRS and U.S. GAAP: it is not amortized but is tested for impairment at least annually. Securities held for trading are reported at fair value with unrealized gains and losses reported on the income statement.

For Further Reference:

(Study Session 9, Module 30.2, LOS 30.e)
CFA® Program Curriculum, Volume 3, page 627

Question #55 of 120

Question ID: 1212052

A U.S. GAAP reporting company holds a number of marketable securities as investments. For the most recent period, the company reports that the market value of its securities held for trading decreased by \$2 million and the market value of its securities available for sale increased in value by \$3 million. Together, these changes in value will:

- A) reduce net income and shareholders' equity by \$2 million.
- B) increase shareholders' equity by \$1 million and have no effect on net income.
- C) **reduce net income by \$2 million and increase shareholders' equity by \$1 million.**

Explanation

Unrealized gains and losses on securities held for trading are included in net income. Unrealized gains and losses on securities available for sale are not reported in net income but are included in comprehensive income. Net income will show a \$2 million loss from the securities held for trading. Shareholders' equity will reflect this loss as well as the \$3 million unrealized gain from securities available for sale, for a net increase of \$1 million.

For Further Reference:

(Study Session 7, Module 21.5, LOS 21.I)

CFA® Program Curriculum, Volume 3, page 113

Question #56 of 120

Question ID: 1207550

Maritza, Inc., is involved in an exchange of debt for equity. In which of the following sections of the cash flow statement would Maritza record this transaction?

- A) Investing activities section.
- B) Financing activities section.
- C) **Footnotes to the cash flow statement.**

Explanation

This transaction results in a reduction of debt and an increase in equity. However, since no cash is involved, it is not reported as a financing activity in the cash flow statement, but will be disclosed in the notes to the cash flow statement.

For Further Reference:

(Study Session 7, Module 23.1, LOS 23.b)

CFA® Program Curriculum, Volume 3, page 187

Question #57 of 120

Question ID: 1212053

Rowlin Corporation, which reports under IFRS, wrote down its inventory of electronic parts last period from its original cost of €28,000 to net realizable value of €25,000. This period, inventory at net realizable value has increased to €30,000. Rowlin should revalue this inventory to:

- A) **€28,000, and report a gain of €3,000 on the income statement.**
- B) €30,000, and report a gain of €3,000 on the income statement.

C) €30,000, and report a gain of €5,000 on the income statement.

Explanation

Under IFRS, inventory values are revalued upward only to the extent they were previously written down. In this case, that is from €25,000 back up to the original value of €28,000. The increase is reported as gain for the period.

For Further Reference:

(Study Session 8, Module 25.4, LOS 25.g)

CFA® Program Curriculum, Volume 3, page 332

Question #58 of 120

Question ID: 1207549

Listed debt securities owned by a company, for which the company intends to collect interest payments and sell the securities, must be carried at fair value with gains and losses reported as other comprehensive income under:

- A)** IFRS only.
- B)** U.S. GAAP only.
- C)** both U.S. GAAP and IFRS.

Explanation

Under IFRS, firms may make an irrevocable choice to carry any financial asset at fair value through profit and loss. Under U.S. GAAP, such debt securities are classified as "available for sale," and carried at fair value with gains and losses reported as other comprehensive income.

For Further Reference:

(Study Session 7, Module 22.3, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

Question #59 of 120

Question ID: 1207562

Which of the following statements regarding an audit and a standard auditor's opinion is *most accurate*?

- A)** The objective of an audit is to enable the auditor to provide an opinion on the numerical accuracy of the financial statements.
- B)** To provide an independent review of a company's financial statements, an external auditor is appointed by the company's management.
- C)** The absence of an explanatory paragraph in the audit report relating to the going concern assumption suggests that there are no serious problems that require a close examination of that assumption by the analyst.

Explanation

A specific explanatory paragraph that makes reference to (questions) the going concern assumption may be a signal of serious problems and call for close examination by the analyst. Therefore, in the absence of such a paragraph, there is no need for a close examination of the going concern assumption by the analyst. The objective of an audit is to enable the auditor to provide an opinion on the fairness and reliability of the financial statements. This is not the same as numerical accuracy. The auditor generally only provides

reasonable assurance that there are no material errors in the financial statements, not an opinion about their numerical accuracy. An external auditor is appointed by the audit committee of the company's board of directors, not by its management. and

For Further Reference:

(Study Session 6, Module 19.2, LOS 19.d)

(Study Session 10, Module 31.1, LOS 31.f)

CFA® Program Curriculum, Volume 3, page 28

CFA® Program Curriculum, Volume 4, page 21

Question #60 of 120

Question ID: 1207557

Two growing firms are identical except that Alfred Company capitalizes costs for some long-lived assets that Canute Company expenses. Alfred is *most likely* to show higher:

- A) net income than Canute.**
- B) working capital than Canute.**
- C) investing cash flow than Canute.**

Explanation

For growing firms, capitalizing results in higher net income compared to expensing. A capitalizing company classifies the costs of the capitalized assets as CFI outflows, while a company that expenses these costs classifies them as CFO outflows. Thus, Alfred's CFO will be higher and his CFI will be lower than Canute's. Working capital is unaffected by the decision to capitalize or expense because the decision does not affect current assets or current liabilities.

For Further Reference:

(Study Session 8, Module 26.1, LOS 26.c)

CFA® Program Curriculum, Volume 3, page 386

Question #61 of 120

Question ID: 1207571

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

Rodgers, Inc., has fixed operating expenses of \$2 million and will break even with sales of \$5 million. For sales of \$7 million, an analyst would estimate the firm's operating income as:

- A) \$800,000.**
- B) \$1,200,000.**
- C) \$2,000,000.**

Explanation

Since at breakeven sales, fixed costs + variable costs = revenue, variable costs for Rodgers at sales of \$5 million must be \$5 million – \$2 million = \$3 million, or 60% of sales. With sales of \$7 million, variable costs are $0.6 \times 7 \text{ million} = 4.2 \text{ million}$. Operating income is then $7 \text{ million} - 4.2 \text{ million} - 2 \text{ million} = \$800,000$.

By assuming some arbitrary price for the product such as \$1,000, this problem can be solved in units as well. Variable costs would be \$600 per unit and operating income would be 7,000 (1,000 – 600) – 2 million = \$800,000.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.e)

CFA® Program Curriculum, Volume 4, page 143

Question #62 of 120

Question ID: 1207567

Weights to be used in calculating a company's weighted average cost of capital are *least appropriately* based on:

- A) information from the company about its target capital structure.
- B) the average capital structure weights for companies of a similar size.**
- C) the average capital structure weights for companies in the same industry.

Explanation

The weights used to calculate WACC should be based on the firm's target capital structure. If the company does not provide information about its target capital structure, an analyst can use the company's current capital structure or the average capital structure weights for the industry. Similar size is not enough for the average weights for other companies to be relevant if those companies are not in the same industry.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.c)

CFA® Program Curriculum, Volume 4, page 80

Question #63 of 120

Question ID: 1207572

An accounts receivable aging schedule is *best* used to:

- A) determine how the receivables turnover ratio has changed over time.
- B) identify trends in how well the firm is doing at collecting receivables and converting them to cash.**
- C) compare a company's receivables management to those of the average for its industry or for a group of peer companies.

Explanation

Information is typically not available to compare the aging of receivables between companies, among groups of companies, or within an industry. Receivables turnover can be calculated from the balance sheet. An aging schedule shows either the absolute or percentage amount of accounts receivable that are current and that are past due by various lengths of time.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.f)

CFA® Program Curriculum, Volume 4, page 175

Question #64 of 120

Question ID: 1207568

The *most appropriate* method to estimate a company's cost of debt capital is to use:

- A) a weighted average of the coupon rates on all the company's outstanding debt.
- B) the yield to maturity on bonds of other companies with the same credit rating as the company's bonds.**
- C) the average rate market rate on outstanding company bonds for a company that issues only floating-rate bonds.

Explanation

A company may use the YTM on the bonds of other companies with the same credit rating and maturity as the company's debt when it is not publicly traded, an example of matrix pricing. Market yields to maturity should be used rather than the coupon rates. For companies that issue floating-rate debt, analysts should use the yield for a longer term that is estimated from the yield curve, rather than using current short-term rates that will likely change over the life of the bonds.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.f)

CFA® Program Curriculum, Volume 4, page 84

Question #65 of 120

Question ID: 1207564

Ron's Organic Markets has limited access to borrowed funds and must choose among ten independent projects with returns greater than their cost of capital. All the projects under consideration have the same required investment of \$2 million, and Ron's has \$10 million available for capital investments this year. Which of the following selection criteria is *least likely* to identify the five projects that will produce the greatest expected increase in the value of the firm? Choose the five projects with:

- A) the highest IRRs.**
- B) the greatest total NPV.
- C) the largest sum of profitability indexes.

Explanation

Since the net present value of the five projects is the expected increase in firm value from undertaking the projects, maximizing the NPV of the projects chosen will result in the selection of the optimal group of five projects. Since the profitability index is the ratio of the present value of the expected after-tax cash flows to the initial outlay, choosing the five projects with the greatest profitability indexes will identify the five projects with the greatest total present values and the projects with the greatest total net present values.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.c)

CFA® Program Curriculum, Volume 4, page 51

Question #66 of 120

Question ID: 1207565

Which of the following is *least* relevant in determining project cash flow for a capital investment?

- A) Sunk costs.**
- B) Tax impacts.**
- C) Opportunity costs.**

Explanation

Sunk costs are not to be included in investment analysis. Opportunity costs and the project's impact on taxes are relevant variables in determining project cash flow for a capital investment.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.b)

CFA® Program Curriculum, Volume 4, page 50

Question #67 of 120

Question ID: 1207573

From a liquidity management perspective, an increase in the number of days of payables is *best* described as:

- A) liquidity neutral.**
- B) a pull on liquidity.**
- C) a source of liquidity.**

Explanation

An increase in the number of days of payables suggests a company is taking longer to pay its vendors. This reduces the cash conversion cycle and represents effective working capital management, a source of liquidity for a company. A decrease in days of payables would be a pull on liquidity because the company is paying its vendors more quickly, which uses cash.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.a)

(Study Session 11, Module 35.1, LOS 35.c)

CFA® Program Curriculum, Volume 4, page 78

Question #68 of 120

Question ID: 1207563

Investing in a project that will help achieve a particular social or environmental objective is *most accurately* described as an example of:

- A) impact investing.**
- B) positive screening.**
- C) responsible investing.**

Explanation

Impact investing refers to promoting a specific social or environmental goal through investment actions, which may include investing in a particular project. Positive screening refers to identifying companies with best practices regarding environmental, social, or governance for consideration when investing. Responsible or sustainable investing refers to integrating social or environmental considerations in general into investment decisions.

For Further Reference:

Question #69 of 120

Question ID: 1207570

When computing weighted average cost of capital (WACC), what is the correct treatment of flotation costs, related to raising additional equity capital?

- A) Increase the discount rate to account for flotation costs.
- B) Adjust the initial project costs by the amount of the flotation costs.**
- C) Flotation costs are not substantial enough to be considered in adjusting the cost of equity.

Explanation

The correct method to account for flotation costs is to make the adjustment in the initial project cost. Adjusting the WACC is incorrect because flotation costs are a cash outflow at the initiation of the project, rather than an ongoing expense. Flotation costs can be substantial, typically 2% to 7% of the amount raised.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.I)

CFA® Program Curriculum, Volume 4, page 105

Question #70 of 120

Question ID: 1212055

An analyst is studying the tax exposures and capital structures of Alpha Corporation and Beta Corporation. Both companies have equivalent weights of debt and equity in their capital structures. Pretax component costs of capital are the same for both companies. Alpha has total capital of \$850 million while Beta has total capital of \$370 million. The marginal tax rates for Alpha and Beta are 35% and 40%, respectively. Which of the following statements regarding Alpha and Beta is *least accurate*?

- A) Beta Corporation has a lower WACC than Alpha Corporation.
- B) An increase in Alpha Corporation's tax rate would decrease its WACC.
- C) A tax rate change will affect Alpha Corporation's cost of equity more than Beta Corporation.**

Explanation

Changes in the tax rate would not affect the cost of equity for either company. If two companies have the same capital structure (i.e., equal weights of debt and equity) and have the same pretax component costs of capital (i.e., equal costs of debt and equity) they will have the same weighted average cost of capital (WACC) only if the companies have the same marginal tax rate. If one company has a higher tax rate, the after-tax cost of debt, $k_d(1 - t)$, will be lower and the WACC, $w_e k_e + w_d k_d(1 - t)$, will be lower as well. Beta Corporation has a lower current WACC since it has a higher tax rate. If Alpha's tax rate increases, its after-tax cost of debt will decrease and its WACC will decrease.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.b)

Question #71 of 120

Question ID: 1207574

A portfolio manager buys \$1 million of U.S. Treasury bills maturing in 90 days at a price of \$990,390 and discount rate of 3.8%. The portfolio also includes the following investments:

- Bank commercial paper maturing in 90 days with a bond equivalent yield of 4.34% and a market value of \$100,000.
- Bank certificates of deposit maturing in six months with a bond equivalent yield of 4.84% and a market value of \$200,000.

The bond-equivalent yield of a comparable benchmark portfolio is 4.0%. Including the Treasury bill purchase, the manager's portfolio is:

- A) outperforming the benchmark.**
- B) underperforming the benchmark.**
- C) performing in line with the benchmark.**

Explanation

The portfolio yield is a weighted average of the yields of the investments that comprise the portfolio. The weights are calculated as the value of each investment relative to the total portfolio value. The bond equivalent yields of the bank investments are given. For the U.S. Treasury bill, the bond-equivalent yield is:

$$(\text{face value} - \text{market value}) / \text{market value} \times 365 / 90 = (\$1,000,000 - 990,390) / 990,390 \times 365 / 90 = 0.0097 \times 0.04056 = 3.93\%.$$

The total market value of the portfolio is \$990,390 + \$100,000 + \$200,000 = \$1,290,390. The portfolio yield equals:

3.93% (\$990,390 / \$1,290,390) + 4.34% (\$100,000 / \$1,290,390) + 4.84% (\$200,000 / \$1,290,390) = 4.10%, which is greater than the yield on the benchmark portfolio.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.e)

CFA® Program Curriculum, Volume 4, page 169

Question #72 of 120

Question ID: 1207566

Langler, Inc., is evaluating two capital projects. Langler has a capital budget of \$50 million. Project P has an internal rate of return of 24% and a net present value of \$5 million. Project Q has an internal rate of return of 18% and a net present value of \$12 million. Project P will cost \$15 million, and Project Q will cost \$48 million. Based on this information, Langler should accept:

- A) Project P to earn the higher return on investment.**
- B) Project Q to maximize shareholder wealth.**
- C) both projects because they both add value to the firm.**

Explanation

When the IRR and NPV methods conflict, the general rule is to take the higher NPV project, as NPV measures the expected increase in the value of the firm from undertaking the project. IRR is not the best measure for ranking mutually exclusive projects of different sizes, and the capital budget is not large enough to do both projects.

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.d, 32.e)

CFA® Program Curriculum, Volume 4, page 52

CFA® Program Curriculum, Volume 4, page 58

Question #73 of 120

Question ID: 1207582

Questions 73 through 85 relate to Equity Investments. (19.5 minutes)

With respect a dividend payment, which of the following dates occurs first?

- A) The record date.
- B) The payment date.
- C) The ex-dividend date.

Explanation

The ex-dividend date occurs prior to the record date and the payment date.

For Further Reference:

(Study Session 13, Module 41.1, LOS 41.d)

CFA® Program Curriculum, Volume 4, page 451

Question #74 of 120

Question ID: 1207580

A company is *most likely* to earn economic profits if it is operating in an industry characterized by:

- A) high industry concentration, high barriers to entry, and low industry capacity.
- B) low industry concentration, low barriers to entry, and low industry capacity.
- C) low industry concentration, high barriers to entry, and high industry capacity.

Explanation

High industry concentration refers to an industry that has a small number of firms, which often leads to less price competition, higher pricing power, and higher return on invested capital. High barriers to entry refer to industries where it is costly for new competitors to enter the industry, which allows companies already in the industry to maintain high profitability and prices. Low industry capacity refers to a situation where demand is greater than supply at current prices, which allows companies to maintain high prices and profits.

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.g)

CFA® Program Curriculum, Volume 4, page 405

Question #75 of 120

Question ID: 1207581

Among the types of factors that influence industry growth and profitability, the one that is *most likely* to affect consumer discretionary goods producers more than consumer staples producers is:

- A) social factors.
- B) demographic factors.
- C) macroeconomic factors.

Explanation

Consumer discretionary goods purchases are very sensitive to economic cycles, while consumer staples are a non-cyclical industry.

For Further Reference:

(Study Session 13, Module 40.1, 40.2, LOS 40.c, 40.j)

CFA® Program Curriculum, Volume 4, page 390

CFA® Program Curriculum, Volume 4, page 422

Question #76 of 120

Question ID: 1207577

Which form of the efficient markets hypothesis (EMH) implies that an investor can achieve positive abnormal returns on average by using technical analysis?

- A) None.
- B) Weak form.
- C) Weak form or semistrong form.

Explanation

An investor cannot achieve positive abnormal returns on average by using technical analysis if prices fully reflect all available security market (price and volume) information. The weak form of the EMH assumes prices reflect this information, and the semistrong and strong forms assume prices reflect additional (non-market) information as well.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.e)

CFA® Program Curriculum, Volume 4, page 330

Question #77 of 120

Question ID: 1207583

A stock has a steady 5% growth rate in dividends. The required rate of return for stocks of this risk class is 15%. The stock is expected to pay a \$1 dividend this coming year. The expected value of the stock at the end of the fourth year is:

- A) \$12.16.
- B) \$14.21.
- C) \$16.32.

Explanation

First solve for D_5 : $D_5 = (D_1)(1 + g)^n = \$1(1.05)^4 = \1.216

$$P_4 = \frac{D_5}{(k-g)} = \frac{\$1.216}{(0.15-0.05)} = \$12.16$$

or

$$P_0 = \frac{1}{0.15-0.05} = 10$$

$$P_4 = 10(1.05)^4 = \$12.16$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #78 of 120

Question ID: 1207575

An investor has long exposure to the risk of the asset underlying an option when taking:

- A) a short position in a put option.**
- B) a short position in a call option.
- C) a long position in a put option.

Explanation

By taking a short position in a put option, the investor has long exposure to the risk in the underlying asset. Because the value of a put option decreases when the price of the underlying asset increases, the value of a short position in a put increases when the price of the underlying increases. Both a short position in a call option and a long position in a put option increase in value when the price of the underlying asset decreases; that is, these option positions have short exposure to the risk of the underlying asset.

For Further Reference:

(Study Session 12, Module 36.2, LOS 36.e)

CFA® Program Curriculum, Volume 4, page 240

Question #79 of 120

Question ID: 1212056

If all other factors remain unchanged, which of the following would *most likely* reduce a company's price/earnings ratio?

- A) The dividend payout ratio increases, and the dividend growth rate increases.
- B) The dividend growth rate increases, and the required rate of return decreases.
- C) The required rate of return increases, and the dividend payout ratio decreases.**

Explanation

$$P/E = \frac{\text{dividend payout ratio}}{k-g}$$

$$g = ROE \times \text{retention rate}$$

Increases in k reduce P/E . Increases in g or the dividend payout ratio increase P/E .

For Further Reference:

(Study Session 13, Modules 41.2, 41.3, LOS 41.g, 41.j)

Question #80 of 120

Question ID: 1207576

Porter, Inc., sells 200,000 newly issued shares to two institutions without registering the shares with its country's securities regulators. This transaction is *best* described as being:

- A) illegal.
- B) in the primary market.**
- C) in the secondary market.

Explanation

Sales of newly issued securities take place in the primary market. Registration of shares sold in private placements of securities is not required. The secondary market refers to the markets in which previously issued securities are traded.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.i)

CFA® Program Curriculum, Volume 4, page 252

Question #81 of 120

Question ID: 1207579

Roger Templeton, CFA, an analyst for Bridgetown Capital Management, is studying past market data to identify risk factors that produce anomalous returns. He tests monthly data on each of 60 financial and economic variables over a 15-year period to find which ones are related to stock index returns. Templeton identifies three variables that show statistically significant relationships with equity returns at a 95% confidence level. What is the *most likely* reason why Bridgetown's management should be skeptical of the anomalies Templeton has identified? The results may suffer from:

- A) data mining bias.**
- B) survivorship bias.
- C) small sample bias.

Explanation

Data mining bias results from the likelihood that some anomalies will show up by chance in a large enough number of tests. A test at the 5% significance level of the hypothesis that stock prices are not correlated with a variable will reject about 1 such hypothesis in 20 when the hypothesis is true.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.f)

CFA® Program Curriculum, Volume 4, page 332

Question #82 of 120

Question ID: 1207586

An asset-based valuation model would *most likely* provide a reliable estimate of market value for:

- A) a pure-play online news site.
- B) a multinational freight distribution firm.
- C) a privately held metal fabrication company.**

Explanation

Asset-based valuation involves determining market values for the subject company's assets and liabilities to calculate the equity value. Compared to the other answer choices, a privately held metal fabrication company will likely have a larger proportion of physical assets relative to intangible assets, and market values for its physical assets should be relatively easier to obtain. An online news site is likely to have a relatively small physical asset base and relatively high intangibles, making this company difficult to value using asset-based valuation. A large multinational corporation is likely to be difficult to value using an asset-based valuation model, even if the company has a large physical asset base relative to its intangibles, because it may have extensive international assets.

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.I)

CFA® Program Curriculum, Volume 4, page 479

Question #83 of 120

Question ID: 1207584

Beachballs, Inc., expects abnormally high earnings for the next three years due to the forecast of unusually hot summers. After the 3-year period, their growth will level off to its normal rate of 6%. Dividends and earnings are expected to grow at 20% for years 1 and 2 and 15% in year 3. The last dividend paid was \$1.00. If an investor requires a 10% return on Beachballs, the price she is willing to pay for the stock is *closest* to:

- A) \$26.00.
- B) \$36.50.**
- C) \$50.00.

Explanation

This is a supernormal growth stock valuation problem.

Step 1: Find the dividends in the supernormal growth period.

$$D_1 = 1.00(1.20) = \$1.20; D_2 = 1.20(1.20) = \$1.44; D_3 = 1.44(1.15) = \$1.656$$

Step 2: Use the constant growth model to find the price at the end of period 2.

$$P_2 = \frac{D_3}{k-g} = \frac{1.656}{0.10-0.06} = \$41.40$$

Step 3: Discount all of the cash flows back to time zero.

$$\begin{aligned} P_0 &= \frac{D_1}{(1+k)} + \frac{D_2}{(1+k)^2} + \frac{P_2}{(1+k)^2} \\ &= \frac{1.20}{1.10} + \frac{1.44}{(1.10)^2} + \frac{41.40}{(1.10)^2} \\ &= \$36.50 \end{aligned}$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #84 of 120

Question ID: 1207578

If stock markets are semistrong-form efficient, a portfolio manager is *least likely* to create value for investors by:

- A) monitoring clients' needs and circumstances.
- B) allocating invested funds among asset classes.
- C) **analyzing financial statements to select undervalued stocks.**

Explanation

Semistrong-form market efficiency implies that fundamental analysis of publicly available information will not generate abnormal returns on average. Portfolio managers should help quantify a client's risk tolerances and return needs, offer portfolio policies and strategies to meet these needs, and construct a portfolio by allocating funds to appropriate asset classes. Portfolio managers can also create value by diversifying their clients' portfolios globally to reduce risk, monitoring and evaluating changing capital market conditions, monitoring their clients' needs and circumstances, and rebalancing their clients' portfolios when necessary.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.e)

CFA® Program Curriculum, Volume 4, page 330

Question #85 of 120

Question ID: 1207601

Jerry Slotz enters an exchange-traded contract that obligates him to purchase a specific amount of an asset on a future date. The contract is *most likely*:

- A) an option contract.
- B) **a futures contract.**
- C) a forward contract.

Explanation

Futures contracts are standardized contracts traded on an exchange that obligate one party to purchase and one party to sell a specific amount of an asset at a specified price on a specified future date. Forward contracts are similar to futures contracts but are customizable (i.e., not standardized) and traded over-the-counter (i.e., not exchange-traded). Call options can be traded on an exchange; however, they give the holder the right, not the obligation, to purchase the asset at a future date.

For Further Reference:

(Study Session 12, Module 36.1, LOS 36.c, Study Session 16, Module 48.1, LOS 48.a, 48.c)

CFA® Program Curriculum, Volume 4, page 218

CFA® Program Curriculum, Volume 5, page 386

CFA® Program Curriculum, Volume 5, page 394

Question #86 of 120

Question ID: 1212058

Questions 86 through 99 relate to Fixed Income. (21 minutes)

Compared to corporate bonds, secondary market trading in government bonds is *most likely* to feature:

- A) brokered markets.
- B) earlier trade settlement.**
- C) narrower bid-ask spreads.

Explanation

Government bond trades typically settle in one day (T + 1) while corporate bond trades typically settle in two or three days (T + 2 or T + 3). Government and corporate bonds trade primarily in dealer markets. Bid-ask spreads depend on an issue's liquidity and may be wider for an illiquid government issue than for a liquid corporate issue.

For Further Reference:

(Study Session 14, Module 43.1, LOS 43.d)

CFA® Program Curriculum: Volume 5, page 70

Question #87 of 120

Question ID: 1212068

An investor buys an option-free bond that has a Macaulay duration of 15.0 and a modified duration of 14.5. If the rate of return on reinvested coupon income is 4.0% and the bond is sold after three years, the investor's annualized holding period return is *most likely* to be:

- A) equal to the bond's yield to maturity at the time of purchase.
- B) less than the bond's yield to maturity at the time of purchase.**
- C) greater than the bond's yield to maturity at the time of purchase.

Explanation

Based on the relationship between Macaulay and modified duration, the bond's yield to maturity at the time of purchase is $(15.0 / 14.5) - 1 = 3.45\%$. If the rate of return on reinvested coupon income is greater than 3.45%, the yield to maturity increased after the investor bought the bond. Over an investment horizon shorter than the Macaulay duration, an increase in YTM decreases the bond's market price by more than it increases reinvestment income. Therefore, the investor's annualized holding period return is less than the yield to maturity at issuance.

For Further Reference:

(Study Session 15, Modules 46.1, 46.3, LOS 46.b, 46.k)

CFA® Program Curriculum: Volume 5, page 248

CFA® Program Curriculum: Volume 5, page 279

Question #88 of 120

Question ID: 1212060

A 10%, 10-year bond is sold to yield 8%. One year passes, and the yield remains unchanged at 8%. Holding all other factors constant, the bond's price during this period will have:

- A) increased.
- B) decreased.**
- C) remained constant.

Explanation

The bond is sold at a premium. As time passes, the bond's price will move toward par. Thus, the price will fall.

$$N = 10; FV = 1,000; PMT = 100; I = 8; CPT \rightarrow PV = \$1,134$$

$$N = 9; FV = 1,000; PMT = 100; I = 8; CPT \rightarrow PV = \$1,125$$

For Further Reference:

(Study Session 14, Module 44.1, LOS 44.b)

CFA® Program Curriculum: Volume 5, page 113

Question #89 of 120

Question ID: 1212057

An analyst is considering a bond for purchase. The bond has a coupon that resets semiannually and is determined by the following formula:

$$\text{coupon} = 12\% - (3.0 \times 6\text{-month Treasury bill rate})$$

This bond is *most* accurately described as:

- A) a step-up note.
- B) an inverse floater.**
- C) an inflation protected security.

Explanation

The bond is an inverse floater because the coupon rate will move opposite to any move in the reference rate.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.e)

CFA® Program Curriculum: Volume 5, page 26

Question #90 of 120

Question ID: 1212065

Asset-backed securities with a lockout period are *most likely* to be backed by:

- A) automobile loans.
- B) home-equity loans.
- C) credit card receivables.**

Explanation

Asset-backed securities backed by credit card receivables have a lockout period, during which principal repayments are reinvested in additional receivables.

For Further Reference:

(Study Session 14, Module 45.2, LOS 45.h)

CFA® Program Curriculum: Volume 5, page 213

Question #91 of 120

Question ID: 1212062

A five-year corporate bond and its benchmark government bond had the following yields over a one-month period:

	Beginning of Month	End of Month
Corporate bond yield	6.75%	7.00%
Government bond yield	4.25%	4.75%

Over this month, the price of the corporate bond *most likely* experienced:

- A) unfavorable macroeconomic and microeconomic factors.
- B) favorable macroeconomic factors and unfavorable microeconomic factors.
- C) unfavorable macroeconomic factors and favorable microeconomic factors.**

Explanation

The benchmark yield increased, which suggests macroeconomic factors were unfavorable for bond prices overall. The corporate bond's spread to its benchmark decreased from 250 basis points to 225 basis points, which suggests microeconomic factors were favorable for the bond's price.

For Further Reference:

(Study Session 14, Module 44.5, LOS 44.k)

CFA® Program Curriculum: Volume 5, page 147

Question #92 of 120

Question ID: 1212066

An investor most concerned with reinvestment risk would be *least likely* to:

- A) prefer a noncallable bond to a callable bond.
- B) prefer a lower coupon bond to a higher coupon bond.
- C) eliminate reinvestment risk by holding a coupon bond until maturity.**

Explanation

The key term here is *coupon bond*. While an investor in a fixed-coupon bond can usually eliminate interest rate risk by holding a bond until maturity, the same is not true for reinvestment risk. The receipt of periodic coupon payments exposes the investor to reinvestment risk. A noncallable bond reduces reinvestment risk by reducing the risk of repayment. Thus, an investor most concerned with reinvestment risk would prefer a noncallable bond to a callable bond. Since lower coupon bonds have lower reinvestment risk, this same investor would prefer a lower coupon bond to a higher coupon bond.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.f, Study Session 15, Module 46.1, LOS 46.a)

CFA® Program Curriculum: Volume 5, page 37

CFA® Program Curriculum: Volume 5, page 241

Question #93 of 120

Question ID: 1212064

For an asset-backed security (ABS), a special purpose entity:

- A) sells an asset to the issuing corporation, which then proceeds to issue the ABS.
- B) is used to separate assets used as collateral from those of the company seeking financing through an ABS.**
- C) acts as an intermediary that purchases an asset from the company issuing an ABS and then resells it to obtain sufficient liquid funds to provide collateral for the ABS.

Explanation

A special purpose entity is a legal corporation to which the assets used as collateral in an ABS issue are sold. This transaction separates the assets backing the ABS from the other assets of the company that creates the SPE.

For Further Reference:

(Study Session 14, Modules 42.1, 45.1, LOS 42.d, 45.b)

CFA® Program Curriculum: Volume 5, page 21

CFA® Program Curriculum: Volume 5, page 181

Question #94 of 120

Question ID: 1212061

The following interest rate information is observed:

Spot Rates	
1 year	10%
2 years	11%
3 years	12%

Based on this data, the 2-year forward rate one year from now is *closest* to:

- A) 12%.
- B) 13%.**
- C) 14%.

Explanation

$${}_2f_1 = \left[\frac{(1+z_3)^{1+2}}{(1+z_1)^1} \right]^{1/2} - 1 = \left[\frac{(1.12)^3}{(1.10)} \right]^{1/2} - 1 = 13.01\%$$

Note: $\frac{(3 \times 12) - (1 \times 10)}{2} = 13$

For Further Reference:

(Study Session 14, Module 44.2, LOS 44.h)

CFA® Program Curriculum: Volume 5, page 143

Question #95 of 120

Question ID: 1212067

A fixed-income portfolio manager is estimating portfolio duration based on the weighted average of the durations of each bond in the portfolio. The manager should calculate duration using:

- A) parallel shifts of the benchmark yield curve.**
- B) equal-sized increases and decreases in a benchmark bond's yield.
- C) equal-sized increases and decreases in the portfolio's cash flow yield.

Explanation

Portfolio duration as a weighted average of the individual bonds' durations is calculated assuming parallel shifts in the yield curve. Cash flow yield is used to calculate duration based on the weighted average time until a bond portfolio's cash flows are scheduled to be received.

For Further Reference:

(Study Session 15, Module 46.2, LOS 46.f)

CFA® Program Curriculum: Volume 5, page 265

Question #96 of 120

Question ID: 1212070

Bert Reed owns a junior secured bond of a firm that has entered bankruptcy proceedings. Mia Tano owns a senior unsecured bond of the same firm. As a result of the bankruptcy, Reed and Tano each recover 80% of the interest and principal owed on their bonds. This outcome is *least likely* the result of:

- A) an order of the bankruptcy court.
- B) the pari passu ranking of these creditors.**
- C) a negotiated settlement among the firm's creditors.

Explanation

Secured creditors have priority of claims over unsecured creditors in a bankruptcy, rather than ranking pari passu with unsecured creditors. However, the priority of claims might not always be followed, either because the creditors have negotiated a different outcome or because a bankruptcy court has ordered one.

For Further Reference:

(Study Session 15, Module 47.1, LOS 47.c)

CFA® Program Curriculum: Volume 5, page 305

Question #97 of 120

Question ID: 1212059

A repurchase agreement is *most accurately* described as:

- A) an embedded option held by a bondholder.
- B) a form of credit enhancement in a bond issue.
- C) a source of short-term funding for a bondholder.**

Explanation

A repurchase agreement is a form of short-term collateralized borrowing in which a bondholder sells a security and agrees to buy it back at a higher price.

For Further Reference:

(Study Session 14, Module 43.2, LOS 43.j)

CFA® Program Curriculum: Volume 5, page 92

Question #98 of 120

Question ID: 1212063

Which of the following is *least likely* a benefit of securitization?

- A) Reducing funding costs.
- B) Removing liabilities from the balance sheet.**
- C) Increasing the liquidity of balance sheet assets.

Explanation

Financial assets are securitized, not liabilities.

For Further Reference:

(Study Session 14, Module 45.1, LOS 45.a)

CFA® Program Curriculum: Volume 5, page 180

Question #99 of 120

Question ID: 1212069

The return impact of a 25 basis point widening in yield for an option-free bond is greatest if it has a duration of:

- A) 4 and convexity of 24.
- B) 5 and convexity of 32.
- C) 6 and convexity of 90.**

Explanation

The impact on return will be greatest for the bond with the highest duration. For a 25 basis point (0.0025) increase in yield, the convexity adjustment is unlikely to be large enough to affect the ordering.

Calculations for each answer choice are as follows:

$$\text{Return impact} = -4(0.0025) + (1/2)(24)(0.0025)^2 = -0.99\%$$

$$\text{Return impact} = -5(0.0025) + (1/2)(32)(0.0025)^2 = -1.24\%$$

$$\text{Return impact} = -6(0.0025) + (1/2)(90)(0.0025)^2 = -1.47\%$$

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.i)

CFA® Program Curriculum: Volume 5, page 270

Question #100 of 120

Question ID: 1207607

Questions 100 through 106 relate to Derivatives. (10.5 minutes)

How does the cost of holding the underlying asset affect option values?

- A) Decreases both call and put values.
- B) Decreases call values and increases put values.
- C) Increases call values and decreases put values.

Explanation

Carrying costs of holding the underlying asset increase the value of call options and decrease the value of put options.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.k)

CFA® Program Curriculum, Volume 5, page 482

Question #101 of 120

Question ID: 1207602

Open interest in a futures market is *most accurately* described as the total number of:

- A) contracts outstanding.
- B) contracts that change hands in a given period.
- C) buyers and sellers currently holding outstanding positions.

Explanation

Open interest is the total number of contracts (long/short position pairs) outstanding. The number of contracts that change hands in a given period is volume, not open interest.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.c)

CFA® Program Curriculum, Volume 5, page 394

Question #102 of 120

Question ID: 1207606

If a plain vanilla interest rate swap is replicated with a series of forward rate agreements (FRAs), at initiation, each FRA must have a forward rate equal to the swap's:

- A) fixed rate and must have a value of zero.
- B) floating rate and must have a value of zero.
- C) fixed rate but may have a non-zero value.**

Explanation

Each FRA would have a forward rate equal to the fixed rate in the interest rate swap but would not necessarily have a value of zero at initiation, although the sum of the values of the replicating FRAs would be zero.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.d)

CFA® Program Curriculum, Volume 5, page 478

Question #103 of 120

Question ID: 1207604

Under which of the following conditions does a forward contract have a positive value to the short party at expiration?

- A) The forward price is less than the value of the forward contract.
- B) The spot price of the underlying asset is less than the forward price.**
- C) The value of the forward contract is less than the spot price of the underlying asset.

Explanation

At expiration, a forward contract has positive value to the short party (and an equal negative value to the long party) if the spot price of the underlying asset is less than the price specified in the forward contract.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.d)

CFA® Program Curriculum, Volume 5, page 469

Question #104 of 120

Question ID: 1212071

At expiration, exercise value is equal to time value for:

- A) an in-the-money call or an out-of-the-money put.
- B) an out-of-the-money call or an in-the-money put.
- C) an out-of-the-money call or an out-of-the-money put.**

Explanation

The time value of an option is zero at expiration. For an out-of-the-money option, the exercise value is zero at expiration.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.j)

Question #105 of 120

Question ID: 1207603

A basis swap may be structured as:

- A) a contingent claim only.
- B) a forward commitment only.**
- C) either a contingent claim or a forward commitment.

Explanation

Interest rate swaps are forward commitments. A basis swap is an interest rate swap in which the parties exchange one floating-rate obligation for another.

For Further Reference:

(Study Session 16, Modules 48.1, 48.2, LOS 48.b, 48.c)

CFA® Program Curriculum, Volume 5, page 387

CFA® Program Curriculum, Volume 5, page 394

Question #106 of 120

Question ID: 1207605

Which of the following is *least likely* to change over the life of a derivatives contract?

- A) The futures price.
- B) The risk-free rate.
- C) The forward price.**

Explanation

The price in a forward contract does not change over the contract's life. The price in a futures contract can change during the life of a futures contract because gains and losses are settled daily to return the value of the contract to zero. The risk-free rate can change at any time.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.b)

CFA® Program Curriculum, Volume 5, page 468

Question #107 of 120

Question ID: 1207611

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

Assume most hedge funds have a 2-and-20 fee structure and most funds of funds have a 1-and-10 fee structure. Over a long investment horizon, compared to net returns from investing directly in hedge funds, net returns from investing in funds of funds are *most likely* to be:

- A) lower.
- B) higher.
- C) the same.

Explanation

Investing in funds of funds is likely to result in lower returns net of fees over time compared to investing directly in hedge funds. Funds of funds receive net-of-fees returns from the funds in which they invest, and charge their own investors an additional layer of management and incentive fees.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.d)

CFA® Program Curriculum, Volume 6, page 20

Question #108 of 120

Question ID: 1212074

The component of the yield on a long commodity futures position that is independent of whether the contract is in contango or backwardation is:

- A) the roll yield.
- B) the collateral yield.
- C) the convenience yield.

Explanation

Collateral yield depends on the yield on T-bills posted as collateral (margin). Roll yield, or the gains and losses that result from entering into a new, longer-dated futures contract as previous contracts expire or are closed out, depends on whether the contract is in contango (futures price greater than spot price) or backwardation (futures price less than spot price). Futures markets for commodities with high convenience yield tend to be in backwardation, while futures markets for commodities with little to no convenience yield tend to be in contango.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

Question #109 of 120

Question ID: 1212072

With respect to venture capital, the term "mezzanine-stage financing" is used to describe the financing:

- A) to prepare for an initial public offering.
- B) to initiate commercial manufacturing.
- C) that supports product development and market research.

Explanation

Mezzanine-stage venture capital financing provides capital during the period prior to an initial public offering.

For Further Reference:

Question #110 of 120

Question ID: 1212075

Cathy Werner, CFA, is considering adding exposure to hedge funds to her portfolio of traditional investments. Based on historical mean and standard deviation of hedge fund index returns and their correlation with traditional investment returns, Werner estimates the diversification benefits from allocating 5% of portfolio assets to hedge funds. Assuming her calculations are correct, Werner is *most likely* to:

- A) overestimate the potential diversification benefits to the portfolio.**
- B) underestimate the potential diversification benefits to the portfolio.
- C) accurately estimate the potential diversification benefits to the portfolio.

Explanation

Historical mean and standard deviation of returns for hedge fund indexes are likely to overestimate expected return and underestimate risk due to survivorship and backfill biases. As a result, determining a portfolio asset allocation to hedge funds based on these measures is likely to overestimate the diversification benefits.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.c, 50.f)

CFA® Program Curriculum, Volume 6, page 57

Question #111 of 120

Question ID: 1212073

A private equity firm that wants to receive money from a portfolio company without giving up control of the portfolio company is *most likely* to engage in:

- A) a trade sale.**
- B) a secondary sale.
- C) a recapitalization.**

Explanation

Recapitalization is when the company issues debt to fund a dividend distribution to equity holders (the fund). It is not an exit, in that the fund still controls the company, but often is a intermediate step toward an exit.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

Question #112 of 120

Question ID: 1207614

A hedge fund has a 2-and-20 fee structure, based on beginning-of-period assets, with a soft hurdle rate of 5%. Incentive fees are calculated before management fees. An endowment invests \$60.0 million in the hedge fund. The value of the endowment's investment, before fees, decreases to \$56.2 million after one year and increases to \$58.0 million the next year. In the second year the endowment will be charged management and incentive fees *closest to*:

- A) \$1.10 million.
- B) \$1.15 million.
- C) **\$1.70 million.**

Explanation

Year 1: Management fee = \$60.0 million $\times$ 2% = \$1.2 million. No incentive fee is charged because the fund decreased in value. Value after fees = \$56.2 million – \$1.2 million = \$55.0 million.

Year 2: Management fee = \$55.0 million $\times$ 2% = \$1.1 million.

Year 2 return = \$58.0 million / \$55.0 million – 1 = 5.45% which is greater than the hurdle rate. With a soft hurdle rate and no high water mark provision, the entire gain is eligible for incentive fees: (\$58.0 million – \$55.0 million) $\times$ 20% = \$0.6 million.

Total fees in Year 2 = \$1.1 million + \$0.6 million = \$1.7 million.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.d)

CFA® Program Curriculum, Volume 6, page 20

Question #113 of 120

Question ID: 1212076

Commodities differ from other alternative investments in that:

- A) **returns are due only to price changes.**
- B) specialized funds are available as an investment vehicle.
- C) they have a low correlation of returns with traditional investments.

Explanation

Unlike alternative asset classes that produce income streams, commodities only generate returns from price changes.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

Question #114 of 120

Question ID: 1207618

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

A security's beta is *best* estimated by the slope of:

- A) the capital market line.
- B) the security market line.
- C) **the security's characteristic line.**

Explanation

Beta, a measure of systematic risk, can be estimated as the slope coefficient from a regression based on the market model, $R_i = \alpha + \beta_i (R_{mkt} - R_f)$. This regression line is the security's characteristic line.

For Further Reference:

Question #115 of 120

Question ID: 1207621

In choosing asset classes for establishing strategic portfolio allocation across assets, the manager would *most* prefer that:

- A) asset classes are only those with tradable liquid assets.
- B) the asset classes span the broadest universe of investable assets.
- C) **correlations of asset returns within an asset class are significantly greater than correlations of asset class returns.**

Explanation

The important points in the determination of asset classes are that assets included in a class have similar performance characteristics and a relatively high correlation of returns between assets, and that returns of asset classes have relatively low correlations to realize the benefits of diversification across asset classes. Some asset classes, such as real estate and hedge funds, may be illiquid. The asset classes in the strategic allocation should be mutually exclusive and cover the universe of investable assets available to the fund manager(s) based on the client's objectives and constraints, not the broadest possible universe of asset classes.

For Further Reference:

(Study Session 19, Module 54.1, LOS 54.f)

CFA® Program Curriculum, Volume 6, page 265

Question #116 of 120

Question ID: 1207622

A bank estimates the expected value of a one-month loss that exceeds ¥100 million to be ¥300 million. The ¥300 million estimate is *best* described as:

- A) a value at risk.
- B) **a conditional VaR.**
- C) a scenario-based VaR.

Explanation

Conditional VaR is the expected value of a loss, given that the loss exceeds a minimum amount. Value at risk is the minimum loss that will occur over a period with a specified probability.

For Further Reference:

(Study Session 19, Module 55.1, LOS 55.g)

CFA® Program Curriculum, Volume 6, page 315

Question #117 of 120

Question ID: 1207616

Given the following correlation matrix, a risk-averse investor would *least* prefer which of the following 2-stock portfolios (all else equal)?

Stock	W	X	Y	Z
W	+1			
X	-0.2	+1		
Y	+0.6	-0.1	+1	
Z	+0.8	-0.3	+0.5	+1

A) W and Y.

B) X and Y.

C) X and Z.

Explanation

A risk-averse investor prefers less risk to more risk. The lower the correlation, the greater the risk reduction. Thus, a risk-averse investor would most prefer the portfolio with the lowest correlation coefficient and least prefer the one with the highest. Of the choices given, W and Y's correlation coefficient of +0.6 is the highest.

For Further Reference:

(Study Session 18, Module 52.2, 52.3, LOS 52.e, 52.g)

CFA® Program Curriculum, Volume 6, page 140

CFA® Program Curriculum, Volume 6, page 152

Question #118 of 120

Question ID: 1207619

Which of the following statements about the security market line (SML) and capital market line (CML) is *most accurate*?

A) The SML involves the concept of a risk-free asset, but the CML does not.

B) The SML uses beta, but the CML uses standard deviation as the risk measure.

C) Both the SML and CML can be used to explain a stock's expected return.

Explanation

The SML and CML both intersect the vertical axis at the risk-free rate. The SML describes the risk/return tradeoff for individual securities or portfolios, whereas the CML describes the risk/return tradeoff of various combinations of the market portfolio and a riskless asset.

For Further Reference:

(Study Session 18, Module 53.1, 53.2 LOS 53.b, 53.f)

CFA® Program Curriculum, Volume 6, page 192

CFA® Program Curriculum, Volume 6, page 213

Question #119 of 120

Question ID: 1207617

Becky Scott and Sid Fiona have the same expectations about the risk and return of the market portfolio; however, Scott selects a portfolio with 30% T-bills and 70% invested in the market portfolio, while Fiona holds a leveraged portfolio, having borrowed to invest

130% of his portfolio equity value in the market portfolio. Regarding their preferences between risk and return and their indifference curves, it is *most likely* that:

- A) Scott is risk averse but Fiona is not.
- B) Fiona's indifference curves are flatter than Scott's.**
- C) Scott is willing to take on more risk to increase her expected portfolio return than Fiona is.

Explanation

Even risk-averse investors will prefer leveraged risky portfolios if the increase in expected return is enough to offset the increase in portfolio risk. Scott's portfolio selection implies that she is more risk averse than Fiona, has steeper indifference curves, and is willing to take on less additional risk for an incremental increase in expected returns than Fiona.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.i)

CFA® Program Curriculum, Volume 6, page 166

Question #120 of 120

Question ID: 1207620

A portfolio manager who is comparing portfolios based on their total risk should *most appropriately* use:

- A) Jensen's alpha.
- B) the Sharpe ratio.**
- C) the Treynor measure.

Explanation

The Sharpe ratio measures excess return per unit of total risk. The Treynor measure and Jensen's alpha are calculated with beta, not standard deviation, and are appropriate for analyzing portfolios based on systematic risk.

For Further Reference:

(Study Session 18, Module 53.2, LOS 53.i)

CFA® Program Curriculum, Volume 6, page 222